

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – ANSWER SCRIPT

Course: Artificial Intelligence (CSA17) | CO3: Knowledge Representation & Reasoning

Case Study 1 – Smart Medical Diagnosis System

Question: A hospital uses a rule-based logical agent for preliminary diagnosis, with rules linking Fever, Cough, Rash and Breathlessness to Flu, Measles and Pneumonia. Facts for Patient A: Fever = True, Cough = True, Breathlessness = True.

(a) Propositional Logic Representation

Symbols used:

```
F = Patient has Fever
C = Patient has Cough
Rs = Patient has Rash
B = Patient has Breathlessness
Flu = Possible Flu
M = Possible Measles
P = Possible Pneumonia
```

Rules:

```
R1: F ^ C -> Flu
R2: F ^ Rs -> M
R3: C ^ B -> P
```

Facts (Patient A): F = True, C = True, B = True, Rs = Unknown

(b) Forward Chaining

```
Step 1: Given facts -> F=T, C=T, B=T
Step 2: Test R1 (F ^ C) -> T ^ T = True => FIRE R1 => derive Flu
Step 3: Test R2 (F ^ Rs) -> Rs unknown => cannot fire
Step 4: Test R3 (C ^ B) -> T ^ T = True => FIRE R3 => derive P (Pneumonia)
```

Final derived diagnoses for Patient A: Possible Flu, Possible Pneumonia.

(c) Backward Chaining – Goal: 'Patient A has Pneumonia' (P)

Goal P depends on rule R3: $C \wedge B \rightarrow P$. To prove P, prove sub-goals C and B.

```
Goal: P
|
Rule fired: R3 (C ^ B -> P)
/ \
Sub-goal: C Sub-goal: B
(fact: True) (fact: True)
| |
MATCHED MATCHED
```

Both sub-goals C and B match known facts (True). Hence both leaves succeed, and the goal **P (Pneumonia)** is confirmed True for Patient A.

Case Study 2 – Autonomous Traffic Management Agent

Question: A FOL knowledge base manages emergency-vehicle routing using rules for ClearPath, GreenSignal/RedSignal and Proceed, with facts about an Ambulance and CarA.

(a) Unification of Rule 1 with the Facts

Rule 1: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$

Step 1: $\text{UNIFY}(\text{Vehicle}(x), \text{Vehicle}(\text{Ambulance}))$
 $\Rightarrow \theta_1 = \{ x / \text{Ambulance} \}$

Step 2: Apply θ_1 to $\text{EmergencyType}(x) \Rightarrow \text{EmergencyType}(\text{Ambulance})$
 $\text{UNIFY}(\text{EmergencyType}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})) \Rightarrow \text{success}$
 θ_1 stays = $\{ x / \text{Ambulance} \}$

Result (MGU): $\theta = \{ x / \text{Ambulance} \}$
Derived fact: $\text{ClearPath}(\text{Ambulance})$

(b) Resolution Proof that $\text{Proceed}(\text{CarA})$ is True

Step 1 – Convert rules to CNF (eliminate implication, i.e. $a \rightarrow b$ becomes $\neg a \vee b$):

Rule1 CNF: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule2 splits into two CNF clauses ($a \wedge b$ consequent):

Rule2a: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Rule2b: $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule3 CNF: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts (unit clauses):

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal: $\neg \text{Proceed}(\text{CarA})$

Resolution steps:

S1: Resolve Rule1{x/Ambulance} with $\text{Vehicle}(\text{Ambulance})$
 $\Rightarrow \neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

S2: Resolve S1 with $\text{EmergencyType}(\text{Ambulance})$
 $\Rightarrow \text{ClearPath}(\text{Ambulance})$

S3: Resolve Rule2a{x/Ambulance} with $\text{ClearPath}(\text{Ambulance})$ (S2)
 $\Rightarrow \text{GreenSignal}(\text{Ambulance})$

S4: Resolve Rule3{x/CarA, y/Ambulance} with $\text{Vehicle}(\text{CarA})$
 $\Rightarrow \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

S5: Resolve S4 with $\text{Behind}(\text{CarA}, \text{Ambulance})$
 $\Rightarrow \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

S6: Resolve S5 with $\text{GreenSignal}(\text{Ambulance})$ (S3)
 $\Rightarrow \text{Proceed}(\text{CarA})$

```
S7: Resolve Proceed(CarA) (S6) with negated goal ~Proceed(CarA)
=> EMPTY CLAUSE (box) -- contradiction => goal proved
```

Since resolving the negated goal with the KB produces the empty clause, **Proceed(CarA) is proved TRUE** by refutation.

(c) Is the KB Sufficient for Two Simultaneous Emergency Vehicles?

No, the current KB is **not sufficient**. Reasoning:

- If two vehicles both satisfy $\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$, Rule 1 and Rule 2 would independently grant $\text{GreenSignal}(x)$ to both, which is unsafe if their paths conflict at the same intersection.
- The KB has no notion of priority, arrival order, or conflicting directions, so it cannot decide which vehicle should get the green signal first.
- **Additional facts needed:** $\text{Priority}(x, \text{level})$, $\text{ArrivalTime}(x, t)$, $\text{ConflictingDirection}(x, y)$.
- **Additional rule needed (example):** for all x, y : $\text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{ConflictingDirection}(x, y) \wedge \text{HigherPriority}(x, y) \rightarrow \text{ClearPath}(x) \wedge \sim \text{ClearPath}(y)$.

This prevents the KB from deriving a logical contradiction (both x and y having GreenSignal on conflicting paths at once).

Case Study 3 – Agricultural Expert Reasoning System

Question: A propositional KB relates SoilDry, IrrigationNeeded, CropWheat, ApplyDripMethod and CropAtRisk. Facts: SoilDry = True, CropWheat = True.

(a) Conversion of P1, P2, P3 and Facts into CNF

P1: SoilDry $\rightarrow$ IrrigationNeeded

```
Eliminate implication ( $a \rightarrow b = \sim a \vee b$ ):  
 $\sim \text{SoilDry} \vee \text{IrrigationNeeded}$  [already CNF]
```

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

```
Eliminate implication:  
 $\sim (\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$   
Move negation inward (De Morgan):  
 $\sim \text{IrrigationNeeded} \vee \sim \text{CropWheat} \vee \text{ApplyDripMethod}$  [CNF]
```

P3: $\sim \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

```
Eliminate implication:  
 $\sim (\sim \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$   
Move negation inward (De Morgan, double negation cancels):  
 $\text{ApplyDripMethod} \vee \sim \text{CropWheat} \vee \text{CropAtRisk}$  [CNF]
```

Facts as unit clauses: SoilDry, CropWheat

(b) Resolution Refutation – Prove ApplyDripMethod

```
Clause set:  
C1:  $\sim \text{SoilDry} \vee \text{IrrigationNeeded}$   
C2:  $\sim \text{IrrigationNeeded} \vee \sim \text{CropWheat} \vee \text{ApplyDripMethod}$   
C3:  $\text{ApplyDripMethod} \vee \sim \text{CropWheat} \vee \text{CropAtRisk}$  (not needed for this proof)  
C4: SoilDry (fact)  
C5: CropWheat (fact)  
Negated goal:  $\sim \text{ApplyDripMethod}$   
  
Step 1: Resolve C1 with C4 (SoilDry)  
=> IrrigationNeeded  
  
Step 2: Resolve C2 with 'IrrigationNeeded' (Step1)  
=>  $\sim \text{CropWheat} \vee \text{ApplyDripMethod}$   
  
Step 3: Resolve Step2 result with C5 (CropWheat)  
=> ApplyDripMethod  
  
Step 4: Resolve 'ApplyDripMethod' (Step3) with negated goal  $\sim \text{ApplyDripMethod}$   
=> EMPTY CLAUSE (box) -- contradiction => theorem proved
```

Resolution proof tree:

```
[] (empty clause)  
/ \  
ApplyDripMethod  $\sim \text{ApplyDripMethod}$  (negated goal)  
/(Step3) \  
 $\sim \text{CropWheat} \vee \text{CropWheat}$  (C5)  
ApplyDripMethod  
/(Step2) \  
 $\sim \text{IrrigationNeeded} \vee \text{IrrigationNeeded}$  (Step1)  
 $\sim \text{CropWheat} \vee /(\text{Step1})$  \  
ApplyDrip (C2)  $\sim \text{SoilDry} \vee \text{SoilDry}$  (C4)
```

IrrigNeeded (C1)

Since the refutation reaches the empty clause, **ApplyDripMethod is proved TRUE.**

(c) Effect of Removing the Fact 'SoilDry'

Without SoilDry as a known fact, clause C1 ($\neg \text{SoilDry} \vee \text{IrrigationNeeded}$) can no longer be resolved to derive IrrigationNeeded – the antecedent is no longer satisfied, so IrrigationNeeded is **not derivable**.

Consequently P2 cannot fire either, so **ApplyDripMethod also becomes non-derivable** (neither proved true nor proved false under classical/open-world resolution).

For P3 ($\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$) to fire, we specifically need $\neg \text{ApplyDripMethod}$ to be an established fact – but 'not proved true' is not the same as 'proved false' in classical logic. So:

- **Under strict classical resolution (open-world assumption):** CropAtRisk **cannot** be derived, because $\neg \text{ApplyDripMethod}$ is never established as a fact.
- **Under the Closed-World Assumption (typical of rule-based expert systems, negation-as-failure):** since ApplyDripMethod cannot be proved, it is treated as False by default, so $\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$ holds and P3 fires, deriving **CropAtRisk = True**.

Conclusion: the answer depends on the reasoning scheme adopted by the expert system; most practical rule-based agricultural advisors use CWA, so in practice CropAtRisk would still be flagged as True once SoilDry (and hence irrigation) is no longer known to have occurred.

Case Study 4 – Wumpus World Knowledge Agent

Question: An agent perceives Stench[1,2], Breeze[1,1], Glitter[2,2] and reasons with rules linking percepts to Wumpus/Pit adjacency, Gold location, and cell safety.

(a) Belief State via Modus Ponens

Percepts: Stench[1,2]=True, Breeze[1,1]=True, Glitter[2,2]=True

MP 1: Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]
Stench[1,2] = True $\Rightarrow$ WumpusAdjacent[1,2] = True

MP 2: Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]
Breeze[1,1] = True $\Rightarrow$ PitAdjacent[1,1] = True

MP 3: Glitter[2,2] $\rightarrow$ Gold[2,2]
Glitter[2,2] = True $\Rightarrow$ Gold[2,2] = True

Derived conclusions: WumpusAdjacent[1,2] is True, PitAdjacent[1,1] is True, Gold[2,2] is True.

(b) Forward Chaining – Locating the Wumpus and Pit

Step 1: WumpusAdjacent[1,2] means Wumpus is in a neighbour of [1,2]:
candidates = { [1,1], [1,3], [2,2] }

Step 2: PitAdjacent[1,1] means Pit is in a neighbour of [1,1]:
candidates = { [1,2], [2,1] }

Step 3: Agent is currently standing safely at [1,1] (start cell), so the Wumpus cannot be at [1,1] (else the agent would already be dead).
 $\Rightarrow$ eliminate [1,1] from Wumpus candidates
Updated Wumpus candidates = { [1,3], [2,2] }

Step 4: [1,1] itself is confirmed Safe (rule 4: $\neg$ Wumpus[1,1] $\wedge$ $\neg$ Pit[1,1] $\rightarrow$ Safe[1,1]).

Result: Wumpus is at [1,3] or [2,2]; Pit is at [1,2] or [2,1] (not yet uniquely determined with current percepts).

(c) Is the Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] Safe?

Apply Rule 4 ($\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]) to each cell on the path:

Cell [1,2]:

- Not a Wumpus candidate (candidates are [1,3], [2,2]) $\Rightarrow$ $\neg$ Wumpus[1,2] holds
 - IS a Pit candidate ({[1,2], [2,1]}) $\Rightarrow$ $\neg$ Pit[1,2] CANNOT be established
- $\Rightarrow$ Safe[1,2] is NOT provable (possible pit)

Cell [2,2]:

- IS a Wumpus candidate ({[1,3], [2,2]}) $\Rightarrow$ $\neg$ Wumpus[2,2] CANNOT be established
 - Glitter[2,2] only confirms Gold, not pit/wumpus absence
- $\Rightarrow$ Safe[2,2] is NOT provable (possible Wumpus)

Conclusion: Since Safe[1,2] and Safe[2,2] cannot both be logically established from the current knowledge (Rule 4's premises are not satisfied for either cell), the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is **NOT proven safe**. It carries a genuine risk of a pit at [1,2] and/or the Wumpus at [2,2]. Before taking this route to collect the gold, the agent should first gather more percepts (e.g. by visiting [2,1] or [1,3] to narrow down the Wumpus/Pit locations) rather than assume safety.